

Hyperbolas

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§1 Isogonal Conjugation

Consider any fixed any triangle ABC . Now, switch every point with its isogonal conjugate. Now observe that isogonal conjugation is an involution(not a homography) in $\mathbb{RP}^2$ which sends lines to circumconics and circumconics to lines. Note that isogonal conjugation is done with the sides AB, BC, CA removed as the isogonal conjugate of a point on a side is not well defined¹.

Problem 1.1. Show that isogonal conjugation preserves cross ratio.

Problem 1.2. Let ABC be a triangle with circumcircle Γ and circumcenter O and ℓ be an arbitrary line(not AB, BC or AC) in $\mathbb{RP}^2$. Now, let $\mathcal{C}$ be the set of isogonal conjugates of all points on ℓ .²

- $\mathcal{C}$ is a circumconic of ABC .
- If ℓ is the line at infinity then $\mathcal{C} = \Gamma$.
- If ℓ lies outside Γ then $\mathcal{C}$ is an ellipse.
- If ℓ is tangent to Γ then $\mathcal{C}$ is a parabola.
- If ℓ intersects Γ twice then $\mathcal{C}$ is a hyperbola.
- If ℓ passes through O then $\mathcal{C}$ is a rectangular hyperbola.³

Lemma 1.3

A circumhyperbola is rectangular iff it contains the orthocenter of the reference triangle.

Proof. Come on, you angle chase this!!

□

Lemma 1.4

Given any point $P \neq H$ in the plane then there is a unique rectangular circumhyperbolas passing through it.

¹limiting arguments work well enough though

²You can use this geogebra applet to see $\mathcal{C}$ as you vary ℓ - <https://www.geogebra.org/m/az2bvaew>

³Both asymptotes must be perpendicular

Proof. A conic is determined uniquely by 5 points and we know that a rectangular circumhyperbola passes through A, B, C, H already so we get a unique rectangular circumhyperbola passing through P . $\square$

Definition 1.5. Let $P \neq H$ be a point in the plane. Now, let $\mathcal{H}$ be the unique rectangular circumhyperbola passing through P . Now, the center of $\mathcal{H}$ is known as the **Poncelet Point** of P with respect to ABC .

Lemma 1.6

The Poncelet point of P lies on the nine point circle of ABC .

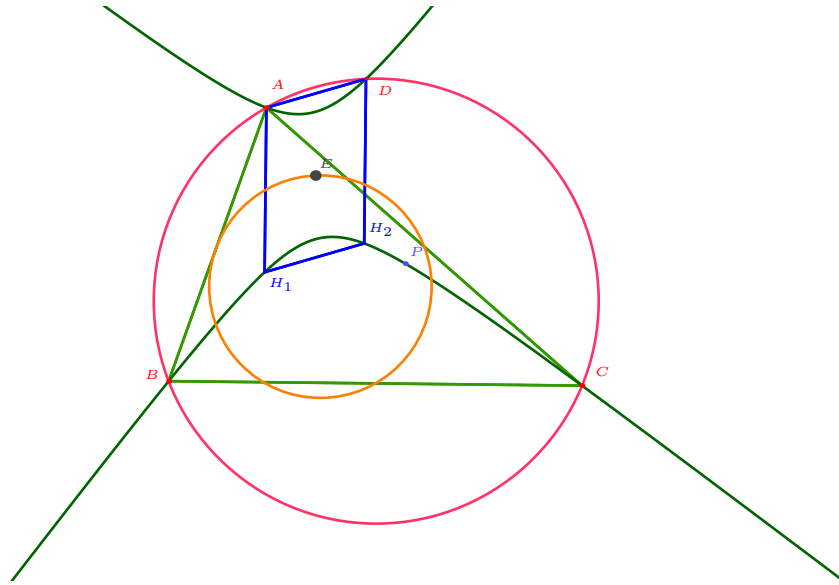


Figure 1: Poncelet Point lies on Nine Point Circle

Proof. Let D be the fourth point at which the hyperbola intersects the circumcircle. Now, we know that the hyperbola passes through the orthocenters of BCA, BCD . Let them be H_1, H_2 . Now, ADH_2H_1 is a parallelogram as $AH_1 \parallel DH_2$ and $AH_1 = DH_2$, thus the center of hyperbola is on midpoint of DH_1 which is on the NPC of ABC . $\square$

Theorem 1.7 (Fundamental Theorem of rectangular circumhyperbolas)

Let $\mathcal{H}$ be a rectangular circumhyperbola of ABC . Let P be a point on $\mathcal{H}$ then the pedal circle of P passes through the Poncelet point of P^a .

^athe center of $\mathcal{H}$

Proof. Problem is equivalent to showing that the pedal circle passes through the intersections of the NPCs of ABC, ABP, ACP, BCP . Let the center of hyperbola be O . Let L, M, N be midpoints of AP, AB, AC respectively. Let D, E, F be the feet of altitudes from P to BC, AC, AB respectively. Now, $\angle EDF = \angle PBA + \angle ACP = \angle LMA + \angle ANL = \angle LMF + \angle ENL = \angle EOF$, thus we are done. $\square$

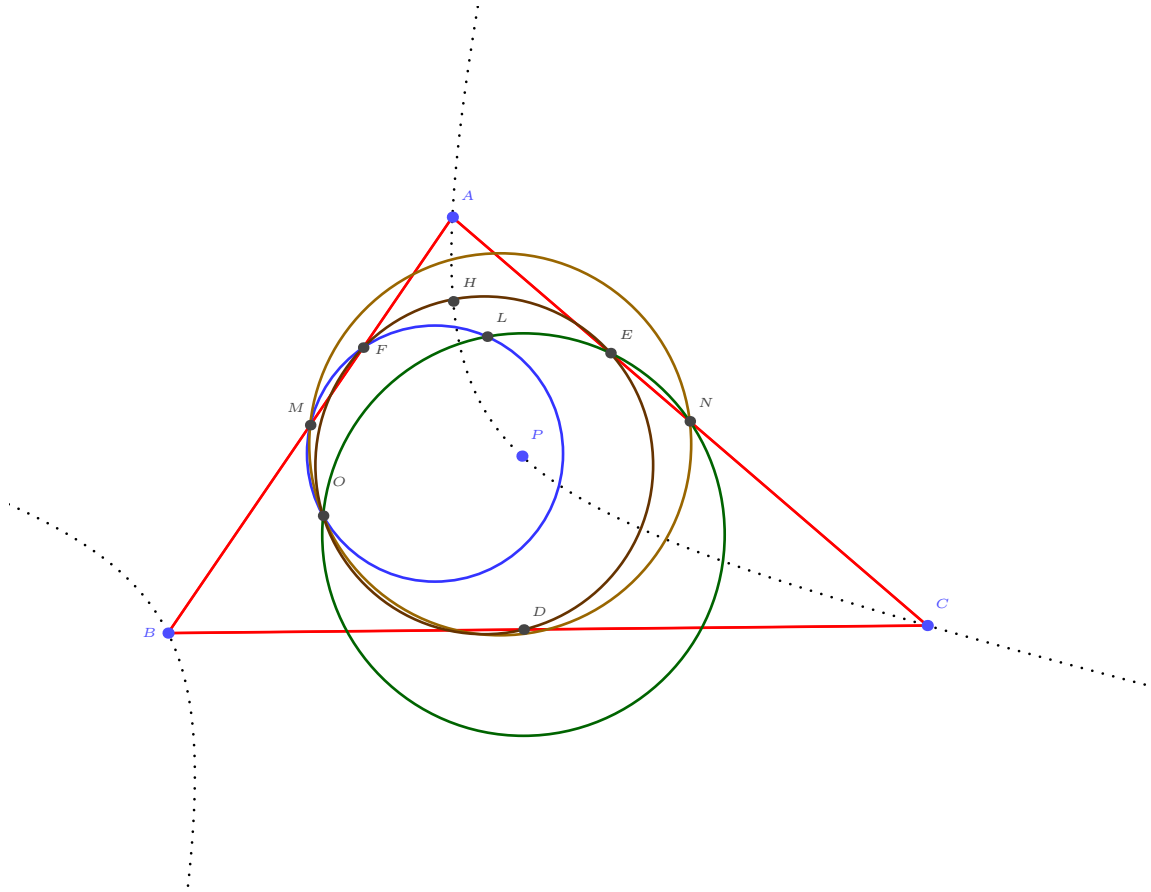


Figure 2: Fundamental Theorem

Theorem 1.8

The cevian circle of P passes through the Poncelet point of P .

There is a very nice proof for this and I will not spoil it for you. But I will provide a walkthrough on how you can achieve the result. If you are unable to still show, you can check the references provided or contact me.

Walkthrough. Consider cevian triangle $A_1B_1C_1$ of P . Let D be any point inside it. Let XYZ be the anticevian triangle of D with respect to $A_1B_1C_1$.

- Show that $ABCDXYZP$ lie on a conic.
- You can do the above by considering appropriate homography or cross-ratio chasing and Pascal's theorem.
- Set D to be the incenter of $A_1B_1C_1$.
- Finish.

□

I hope you were able to develop some intuition for the topic and liked it and the powerful lemmas. Due to lack of time, I could not put example problems in this section and may do that at a later stage.

§2 Problems

The problems in this section may not necessarily be related to Hyperbolas and you may not need to use their properties but simply establish properties by angle chasing. Some problems may be used to build ideas for later problems and in themselves not be related to hyperbolas.

Many others and I have found that angle chasing is extremely powerful in these problems due to the abundance of circles produced by Isogonal Conjugation and considering Poncelet Points. Cross ratio chasing, homographies and other projective geometry ideas are also quite helpful. Moving points are also extremely useful while working on these problems.

§2.1 Stuff you should know

Problem 2.1 (Six Point Circle). Let ABC be a triangle. Now, let P be any point in the plane. Now, let P' be the isogonal conjugate of P . Let $A_1B_1C_1$ be the pedal triangle of P and $A_2B_2C_2$ is the pedal triangle of P' . Then, $A_1A_2B_1B_2C_1C_2$ is a cyclic hexagon with center as the midpoint of PP' .

Problem 2.2 (Anti-Steiner Point). Let ABC be a triangle with orthocenter H . Let ℓ be any line through H then the reflections of ℓ over AB, BC, AC concur on (ABC) .

Problem 2.3 (Isogonality Lemma). Let AP, AS and AQ, AR be two pairs of isogonal lines with respect to $\angle BAC$. Let $PR \cap QS = X$ and $PQ \cap RS = Y$. Then AX, AY are isogonal line with respect to $\angle BAC$.

§2.2 Practice problems

Problem 2.4. Let ABC be a triangle with orthocenter H . Let D be a point on the circumcircle. The Poncelet point of D is the midpoint of HD .

Problem 2.5 (ELMO 2012/5). Let ABC be an acute triangle with $AB < AC$, and let D and E be points on side BC such that $BD = CE$ and D lies between B and E . Suppose there exists a point P inside ABC such that $PD \parallel AE$ and $\angle PAB = \angle EAC$. Prove that $\angle PBA = \angle PCA$.

Problem 2.6 (Second Fontené). If a point P moves on the fixed line ℓ which passes through the circumcenter O of triangle ABC then the pedal circle of P with respect to triangle ABC intersects the Nine-point circle of triangle ABC at a fixed point.

Problem 2.7 (Third Fontené). Denote the isogonal conjugate of P with respect to triangle ABC as P' . Then the pedal circle of P is tangent to the Nine-point circle of triangle ABC if and only if O, P, P' are collinear.

Problem 2.8 (Antigonal Conjugate). Let P be any point other than the orthocenter of ABC . Its antigonal conjugate Q is the reflection of P across the center of the rectangular circumhyperbola of ABC passing through P . Prove that-

- We have $\angle BPC = -\angle BQC$
- The isogonal conjugates of P, Q are inverses of each other in (ABC) .

Problem 2.9 (AoPS). Consider a circle ω with diameter $\overline{KL}$, and let B, C be inverse points on line KL . Fix a point A outside BC . For any $P \in \omega$, let Q be the second intersection of AP and ω , and define $T = KP \cap LQ$.

Prove that as $P \in \omega$ varies, the locus of T is a rectangular hyperbola.

Problem 2.10. Let $\mathcal{H}$ be a rectangular hyperbola with center Z . Let PQ be a chord of $\mathcal{H}$ with midpoint M . Show that the pole of PQ lies on ZM and that the asymptotes of $\mathcal{H}$ are the angle bisectors of line ZM and the line through Z parallel to PQ .

Problem 2.11 (AoPS). Let ABC be a triangle with orthocenter H and let ω be a circle in the plane with center O . If $\mathcal{H}$ is the hyperbola that passes through A, B, C, H, O and A', B', C' are the poles of $\overline{BC}, \overline{CA}$ and $\overline{AB}$ wrt. ω , prove that $\overline{AA'}, \overline{BB'}, \overline{CC'}$ are concurrent in a point that lies on $\mathcal{H}$.

§2.3 First Fontené and Corollaries

Problem 2.12 (First Fontené). Let $\triangle ABC$ have medial triangle $A_1B_1C_1$. Let $P \neq H$ be a point in the plane with pedal triangle $A_2B_2C_2$. Then, let $B_1C_1 \cap B_2C_2 = A_3$, $A_1C_1 \cap A_2C_2 = B_3$, $A_1B_1 \cap A_2B_2 = C_3$. Then, we have that A_2A_3, B_2B_3, C_2C_3 concur on the pedal circle of P .

Problem 2.13. Carry over notation from 2.12. Show that the orthocenter of $A_3B_3C_3$ is the circumcenter of $A_2B_2C_2$.

Problem 2.14. Carry over notation from 2.12. Let Q be the concurrency point we got. Let P' be the isogonal conjugate of P . Let $A_4B_4C_4$ be the pedal triangle of P' . Prove that the simpson line of Q with respect to $A_4B_4C_4$ is parallel to the simpson line of Q with respect to $A_1B_1C_1$.

§2.4 Feurbach

§2.4.1 Buildup

Problem 2.15 (Gergonne Point). Let ABC be a triangle with A, B, C intouch points D, E, F respectively. Then show that AD, BE, CF are concurrent. Call the point of concurrency as the Gergonne point of ABC .

Problem 2.16 (Nagel Point). Let ABC be a triangle, now let A_1, B_1, C_1 be the A, B, C extouch points. Show that AA_1, BB_1, CC_1 concur. Call the point of concurrency as the Nagel Point of ABC .

Problem 2.17 (Feurbach Point). Show that the incircle and nine point circles of $\triangle ABC$ are tangent. Call this point the Feurbach Point of ABC .

Problem 2.18 (Mittenpunkt). Let I_A, I_B, I_C be the A, B, C excenters of ABC . Let L, M, N be midpoints of BC, AC, AB respectively. Prove that LI_A, MI_B, NI_C concur at the symmedian point of $I_AI_BI_C$. Call this point the Mittenpunkt.

Problem 2.19 (Schiffler Point). Let I be the incenter of $\triangle ABC$. Show that the Euler lines of BIC, AIC, AIB, ABC concur. Call this point the Schiffler point.

§2.4.2 The magic hyperbola

Problem 2.20 (Feurbach Hyperbola). Let ABC be a triangle with incenter I . Let $\mathcal{H}$ be the rectangular circumhyperbola of ABC passing through I . Show that the Orthocenter, Gergonne, Nagel, Mittenpunkt and Schiffler of $\triangle ABC$ lie on $\mathcal{H}$ and $\mathcal{H}$ is centered at the Feurbach Point of ABC .

§2.5 Harder problems

Problem 2.21. Let ABC be a triangle and P be a point in the plane. Let $A_1B_1C_1$ be the medial triangle of ABC and $A_2B_2C_2$ be the pedal triangle of P . Let A_3 be reflection of A_2 in A_1 . Similarly define B_3, C_3 . Show that $(A_1B_1C_1), (A_2B_2C_2), (A_3B_3C_3)$ are concurrent.

Problem 2.22 (Micheal Ren). Let ABC be a triangle with Feurbach point Fe , medial triangle MNP , and intouch triangle DEF . Let $EF \cap NP = X$, $FD \cap PM = Y$, and $DE \cap MN = Z$. Let ℓ be the common tangent of the incircle and nine-point-circle.

- Prove that DX, EY , and FZ concur at Fe .
- Prove that $AX \parallel BY \parallel CZ$.
- Prove that $A \in YZ, B \in ZX$, and $C \in XY$.
- Prove that $BC \cap YZ, CA \cap ZX, AB \cap XY \in \ell$.
- Prove that MX, NY , and PZ concur on ℓ .

§3 References

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